Differential Equations Workbook

Series solutions

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POWER SERIES BASICS

■ 1. Where is the power series centered?

$$\sum_{n=0}^{\infty} c_n(x+3)^n = c_0 + c_1(x+3) + c_2(x+3)^2 + c_3(x+3)^3 + \ldots$$

■ 2. Find the radius of convergence of the power series.

$$\sum_{n=0}^{\infty} \frac{3^{2n}}{100^n}(10x)^n$$

■ 3. Find the Maclaurin series representation of $f(x) = \cosh(9x^2)$.

■ 4. Find the Maclaurin series representation of the function.

$$f(x) = 4\ln(-2 - 3x^3)$$

■ 5. Find the Maclaurin series representation of the function.

$$f(x) = \frac{6}{\sqrt{2x} - 1}$$

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6. Find the first three derivatives of the power series.

$$f(x) = \sum_{n=0}^{\infty} 3(-1)^n (2x^2)^n$$

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ADDING POWER SERIES

■ 1. Are the power series in phase? Do the indices match?

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{9^n} x^{2n}$$

■ 2. As written, can the power series be added?

$$\sum_{n=0}^{\infty} n!x^n \qquad 5 + \sum_{n=1}^{\infty} \frac{3^n}{n!} x^n$$

■ 3. Find the sum.

$$\sum_{n=3}^{\infty} (2n-1)c_n x^{n-3} + \sum_{n=1}^{\infty} 2nc_n x^{n-1}$$

■ 4. Add the power series.

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} \qquad x\sum_{n=0}^{\infty} c_n x^n$$

■ 5. Find the sum.

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$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - x\sum_{n=0}^{\infty} c_n x^n$$

6. Find the sum.

$$3x^2\sum_{n=0}^{\infty}(n-2)(n-1)nc_n x^n - (x^2-2)\sum_{n=1}^{\infty} c_n x^{n+1} + (x+1)\sum_{n=2}^{\infty} n(n+1)c_n x^{n-1}$$

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POWER SERIES SOLUTIONS

■ 1. Is $x_0 = 4$ an ordinary point of the differential equation?

$$\frac{1}{x-4}y'' + (x^2+2)y' + 3y = 0$$

■ 2. Can we find a power series solution to the differential equation around $x_0 = -1$?

$$(x-1)y'' + xy' + (x^2+2)y = \sqrt{x-1}$$

■ 3. Find the minimum radius of convergence of a power series solution of the differential equation about the ordinary point $x_0 = -2$.

$$x^3y'' + (2x+1)y' - y = 0$$

■ 4. Find a power series solution to the differential equation, given $y(0) = 0$ and $y'(0) = 1$.

$$(x^2+1)y'' + 2xy' = 0$$

■ 5. Find a power series solution to the differential equation, given $y(0) = 1$ and $y'(0) = 0$.

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$$y'' - xy' - y = 0$$

■ 6. Find a power series solution to the differential equation, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + x^2y' + xy = 0$$

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NONPOLYNOMIAL COEFFICIENTS

■ 1. Solve the differential equation around $x_0 = 0$.

$$y'' + \frac{1}{1-x}y' = 0$$

■ 2. Find the first six terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' - e^x y = 0$$

■ 3. Find the first five terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y' + e^{3x}y = 0$$

■ 4. Find the first eight terms of the power series solution of the differential equation around the ordinary point $x_0 = 0$.

$$y'' + y\cos x = 0$$

■ 5. Find the first six terms of the power series solution of the differential equation around the ordinary point $x_0 = 0$.

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$$y'' - \ln(1 + 5x)y = 0$$

■ 6. Find the first four terms of the power series solution of the differential equation around the ordinary point $x_0 = 2$.

$$y'' + \cosh(x - 2)y = 0$$

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SINGULAR POINTS AND FROBENIUS' THEOREM

■ 1. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$2x^2y'' + x(1 - 2x)y' + 4xy = 0$$

■ 2. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$4xy'' + 2y' + y = 0$$

■ 3. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$x^2y'' + x^2y' - 6y = 0$$

■ 4. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

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$$8x^2y'' + x^2y' + (2 - x)y = 0$$

■ 5. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$3x^2y'' + 3xy' - 2xy = 0$$

■ 6. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$3xy'' + 2y' + 2y = 0$$

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